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Voltage Follower, Part 1

Welcome to the 33rd article in the "Circuit Intuitions" column series. As the title suggests, each article provides insights and intuitions into circuit design and analysis. These articles are aimed at undergraduate students but may serve the interests of other readers as well. If you read this article, I would appreciate your comments and feedback as well as your requests and suggestions for future articles in this series. Please email me your comments at ali@ece.utoronto.ca.

As the name suggests, a voltage follower, also known as a *voltage buffer*, produces an output voltage waveform that closely follows the input voltage waveform. An ideal voltage follower has four important characteristics: an infinite input impedance, a zero output impedance, a no-load voltage gain of 1 V/V, and an infinite bandwidth. The infinite input impedance enables the buffer to not sink any current from the input signal source; otherwise, it may cause a drop in the signal voltage at the buffer input. The zero output impedance enables the ideal buffer to provide any current to the load without any voltage drop. A voltage gain of one will replicate the input voltage at the output node, and its infinite bandwidth guarantees that this voltage gain will remain fixed at 1 V/V irrespective of the frequency. An actual implementation of a voltage follower does not usually possess one or more of these ideal properties, as we discuss in this article. For simplicity, we first provide our analysis of the circuits at low frequencies; then, we move to high frequencies.

A simplified model of an ideal voltage buffer is shown in Figure 1, where R_{in} and R_{out} represent the input and output resistances, respectively, and V_{oc} represents the open-circuit (no-load) voltage at the output of the buffer (for a voltage gain of 1 V/V). A common implementation of a voltage follower consists of an NMOS transistor with its drain tied to the power supply and its source to a bias current, as shown in Figure 2(a). Another implementation consists of a PMOS transistor with its drain tied to the ground and its source to a bias current, as shown in Figure 2(b). In both implementations, the input voltage is applied to the gate of the transistor, and the output is taken from the source of the transistor. Let us examine these circuits to see how closely they mimic an ideal voltage follower. To make our analysis easier, we assume the transistors are in the saturation region, and we initially ignore the body effect and the channel-length modulation; i.e., assume $g_{me} = g_m + g_{mb} = g_m$, and assume $1/g_m \ll r_o$, where g_m and g_{mb} are the short circuit transconductance from the gate and the body of the transistor, respectively; g_{me} is the sum of the two, and r_o is the output resistance of the transistor.

Both circuits exhibit infinite input resistance simply because the gate of an MOS transistor is isolated from its channel via a dielectric, which is an insulator. Also, under no-load conditions, the current through the transistor is fixed at I_B, implying a constant voltage (i.e., dc) across the gate to the source of the transistor. As a result, the voltage waveforms at the gate and the source terminals are expected to be identical except for a dc voltage shift. Ignoring the dc component, the ac signal at the source terminal is expected to follow the ac signal at the gate—hence, the name *source follower* for both implementations.

The difference between using NMOS and PMOS is in the dc level shift that the output voltage experiences. Whereas, in the NMOS case, the output

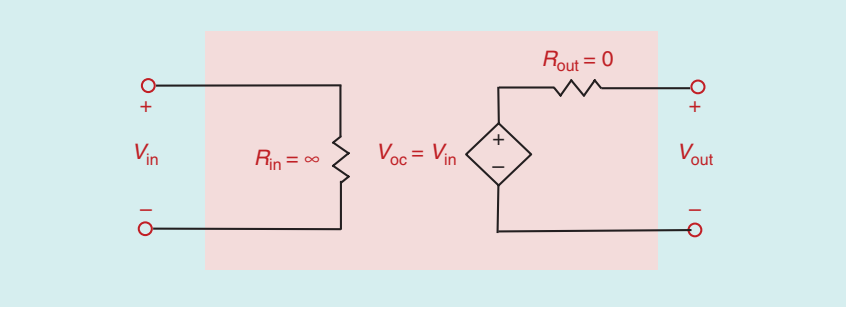


FIGURE 1: A circuit model for an ideal voltage buffer.

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dc voltage is a V_{GS} (>0) lower than the input dc voltage, in the PMOS case, the output dc voltage is a V_{SG} (>0) higher than the input dc voltage. However, in both cases, the open-circuit voltage gain for ac signals is expected to be 1 V/V, which is the same as that of an ideal voltage follower.

If we do not ignore the body effect and the channel-length modulation, the small-signal voltage gain of the source follower will be somewhat lower than 1 V/V. In fact, using the techniques presented in the first article in this series [1], one can easily show that

$$\frac{V_{out}}{V_{in}} = \frac{g_m}{g_{me} + \frac{1}{r_o}} \quad (1)$$

Since g_m is always smaller than g_{me}, this voltage gain is always less than 1 V/V.

Figure 3 shows simulated voltage waveforms for an NMOS and a PMOS source follower designed in a generic 45-nm CMOS technology, with the specific parameters shown in Table 1. In the case of the NMOS, the dc voltage of the output waveform is around 520 mV lower than that of the input waveform. This voltage difference is, in fact, the V_{GS} used to bias the transistor. The peak-to-peak value of the output is a fraction (~ 0.7) of the peak-to-peak value of the input waveform. This reduction in voltage gain (from an ideal of one) is the direct consequence of the body effect and the finite output resistance of the transistor, as listed in Table 1. In fact, by plugging the values from Table 1 into (1), we get the same voltage gain as obtained from the simulation. Readers are encouraged to make sense of the waveforms for the PMOS source follower.

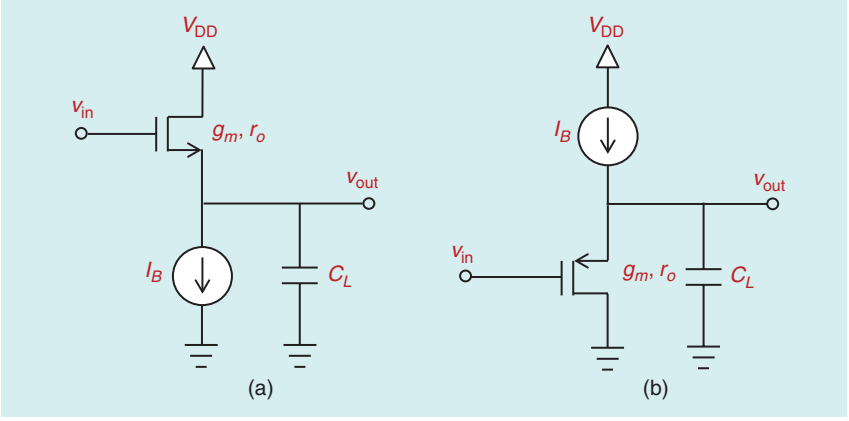


FIGURE 2: Two implementations of a voltage buffer, also known as a *source follower*, using (a) an NMOS transistor and (b) a PMOS transistor.

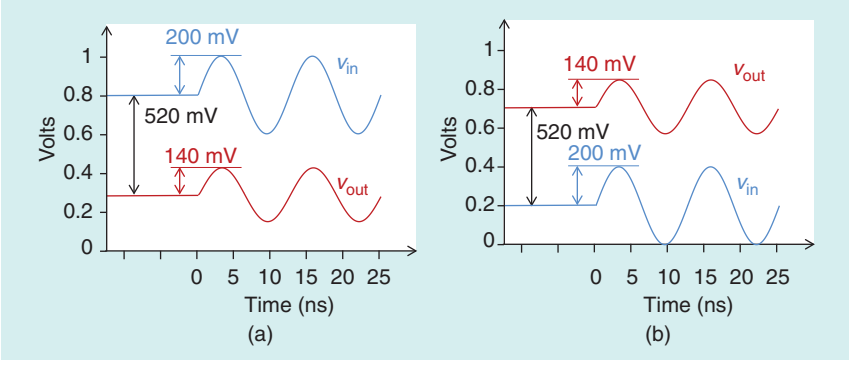


FIGURE 3: A pair of input and output waveforms of (a) an NMOS source follower and (b) a PMOS source follower.

TABLE 1. THE NMOS AND PMOS PARAMETERS CORRESPONDING TO TWO SIMULATED SOURCE FOLLOWERS.

PARAMETER	NMOS	PMOS
W/L	400 nm/45 nm	1,150 nm/45 nm
$\lvert V_T \rvert$	0.43 V	0.35 V
g_m	0.5 mS	0.47 mS
g_{mb}	0.15 mS	0.13 mS
r_o	11.15 kΩ	9.5 kΩ

The W/L values are chosen such that the two transistors have a comparable g_m.

To find the output resistance of the source follower, we look into the source of the transistor (while its gate is grounded), and we see a small-signal resistance of $1/g_{me} \| r_o$ [1]. This resistance, which is around 1.4 kΩ for our two circuits, based on the numbers in Table 1, is certainly a considerable deviation from the ideal value of zero. However, this deviation finds relevance only in relation to the load resistance. If the load resistance is much larger than 1.4 kΩ, then our source follower and the ideal voltage buffer would act similarly. In a sense, the output resistance of the source follower determines a lower bound on the load resistance, below which the deviation from the ideal voltage buffer would become unacceptable. In our case, for load resistances above 14 kΩ, the voltage gain will never drop below 0.9 V/V, and this may be acceptable to us. However, if we wish to drive a load resistance of 1.4 kΩ or less, the voltage gain will drop to below 0.5 V/V, which may be unacceptable.

Figure 4 shows the dc characteristics of the two source followers and compares them against that of an ideal voltage follower. We distinguish two clear deviations here:

- The range of the input voltage over which either the NMOS or the PMOS source follower operates properly

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is limited to around 0.6 V in our example. The NMOS source follower operates properly when the input voltage is in the 0.4–1-V range, whereas the PMOS source follower operates properly in the 0–0.6-V input voltage range. These are both in contrast to the 0–1-V range that we expect from an ideal voltage follower.

- The voltage gain of the two source followers (in their corresponding regions) is about 0.7 V/V, a clear deviation from the ideal gain of 1 V/V.

When the load is a capacitor, there is another interesting behavior in response to a step voltage at the input. To see this behavior, assume that the NMOS source follower of Figure 1(a) is initially biased in the saturation region. Suppose we now apply a positive step to the gate, i.e., increase the dc voltage of the gate by some positive value. In that case, the NMOS transistor will momentarily experience an increase in its v_{GS} simply because its source voltage cannot change instantaneously, as it is connected to the load capacitance. As a result, a transient current will be supplied by the transistor to the capacitor, increasing the capacitor voltage gradually toward its steady state, which is, again, a V_{GS} below the new gate voltage.

A larger step size will result in a larger transient current, increasing the rate of change of the output voltage, also known as the *slew rate*. A negative step voltage, on the other hand, can momentarily bring the v_{GS} to below the transistor threshold, pushing it into the cutoff region, before returning to the steady state and conducting the bias current I_B. This is, again, because the voltage across the capacitor cannot be changed instantaneously.

Figure 5 shows the transient simulation results corresponding to this behavior for our example NMOS source follower. A sudden rise in the input (shown in blue) results in a surge in v_{GS} (shown in black) and a corresponding increase in the transistor current that will gradually charge the load capacitor. As the capacitor voltage rises, v_{GS} will decay to its steady state. This response follows the behavior of an *RC* circuit. When the input voltage suddenly falls, v_{GS} falls below the black dashed line, which marks the transistor threshold voltage. While v_{GS} is below the threshold, the only source of current connected to the load capacitance is the bias current, which depletes the capacitor charge at a constant rate of I_B. Equivalently, the output voltage will drop at a constant slew rate of I_B/C_L.

Our main observation here is that there is a different slew rate for the rising and falling edges of the output voltage. While the positive slew rate will increase with the positive voltage, the negative slew rate is limited by the bias current provided. This asymmetry in the slew rate is another deviation from the ideal voltage follower behavior.

Finally, let us find the bandwidth of a source follower. Assuming that the time constant associated with the load capacitance is dominant—i.e., it is much larger than all other time constants in this circuit—the bandwidth is simply $1/2\pi R_{out} C_L$. For the two circuits simulated in this article, this bandwidth is calculated by hand to be around 1.15 GHz and 1.06 GHz for the NMOS and PMOS source followers, respectively. These hand calculations are within 1% of the results obtained by simulation. To increase the bandwidth, for a constant load capacitance, we must reduce R_{out}, which is a strong function of g_m. However, increasing g_m comes at the cost of increasing the power consumption and the area of the source follower. An important question here is whether we can modify the design of the voltage follower to reduce the output resistance and, hence, increase the bandwidth, without increasing the power consumption or the area. We will review other voltage follower topologies in the next article in this series.

In summary, the simple source follower comes close to achieving two of the ideal characteristics: it exhibits an infinite input resistance and an open circuit voltage gain of only slightly below 1 V/V. Its output resistance, however, is on the order of $1/g_m$, which may not be low enough for certain loads and is certainly not low enough for capacitive loads at higher frequencies. Nevertheless, this simple source follower is a good first attempt at producing a voltage follower.

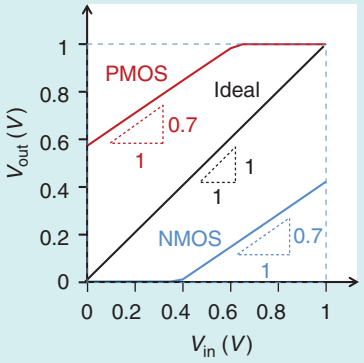


FIGURE 4: The simulated dc behavior of NMOS (in blue) and PMOS (in red) source followers compared to that of an ideal voltage follower (in black).

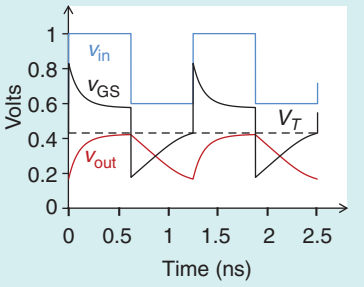


FIGURE 5: An input waveform (in blue) and its corresponding simulated output waveform (in red) for the NMOS source follower reveal an asymmetry in the slew rate between the rising and falling edges. The curve in solid black represents v_{GS} (= v_{in} – v_{out}).

Acknowledgment

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Reference

[1] A. Sheikholeslami, "Looking into a node [Circuit Intuitions]," *IEEE Solid-State Circuits Mag.*, vol. 6, no. 2, pp. 8–10, Spring 2014, doi: 10.1109/MSSC.2014.2315062.

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